

10-Mark Questions & Answers — Data Structures Module 4: Trees

Q1. Explain Binary Trees, Tree Traversals, BST operations, and AVL Tree balancing in detail with diagrams and C implementations.

Comprehensive explanation covering:

1. Binary Tree definitions, array and pointer representations.
2. Inorder, Preorder, Postorder, and Level Order traversal algorithms.
3. BST Search, Insertion, and Deletion algorithms.
4. AVL Tree Balance Factor, LL/RR/LR/RL rotation cases, and step-by-step rebalancing trace.
5. Introduction to Multi-way B-Trees, B+ Trees, and Red-Black Trees.